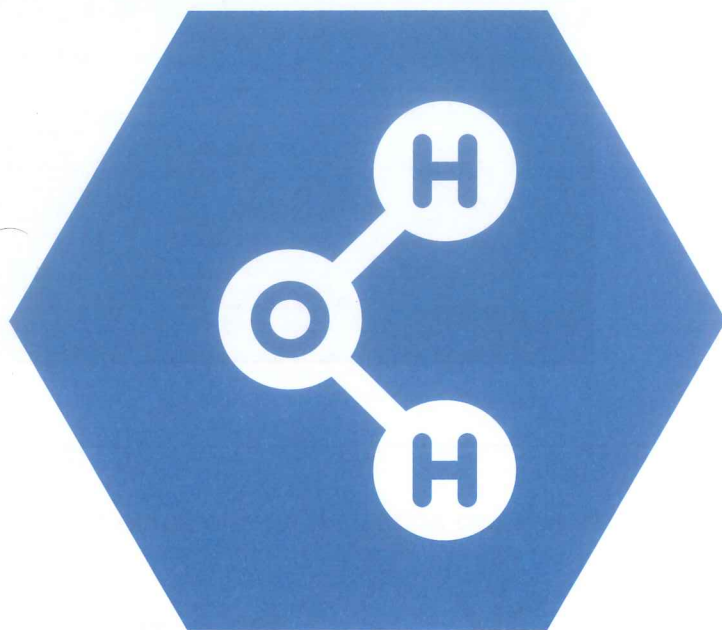




UNSW Global
AUSTRALIA

INTRODUCTORY PAPER



2015 ICAS

International Competitions
and Assessments for Schools

SCIENCE

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

30 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a 2B or B pencil.

Do **NOT** use a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

There are **30 MULTIPLE-CHOICE QUESTIONS** (1–30).

Use the information provided to choose the **BEST** answer from the four possible options.

On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a calculator and a ruler.

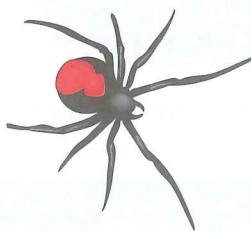
1. The photograph shows some living and non-living things in a fish tank.



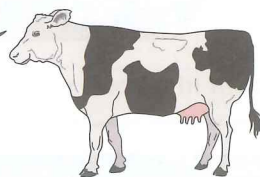
Which is a living thing?

- (A) plant
- (B) water
- (C) rock
- (D) sand

2. Which two animals have the same number of legs?



spider



cow



snake



cicada



otter

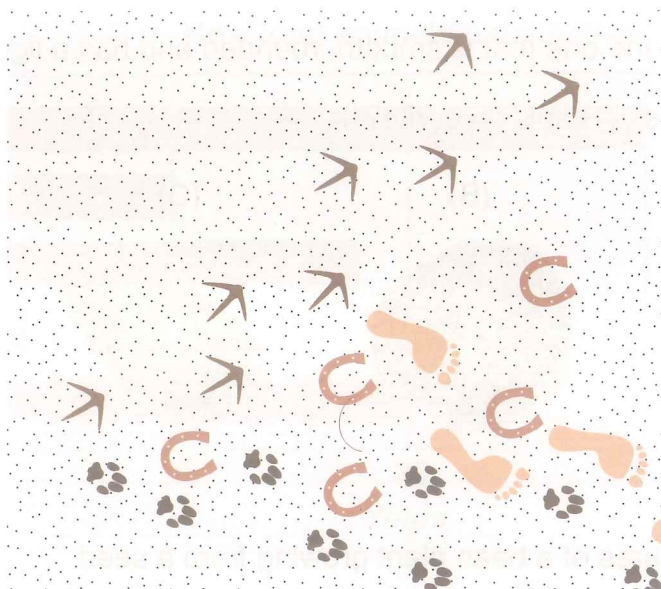
- (A) spider and cow
- (B) snake and otter
- (C) cow and otter
- (D) spider and cicada

3. What does this sign mean?



- (A) Do not smoke here.
- (B) Buy your cigarettes here.
- (C) Cigarettes are good for you.
- (D) Smoking is allowed here.

4. These footprints were made in the sand.



Which footprints were made by a bird?

(A)



(B)



(C)

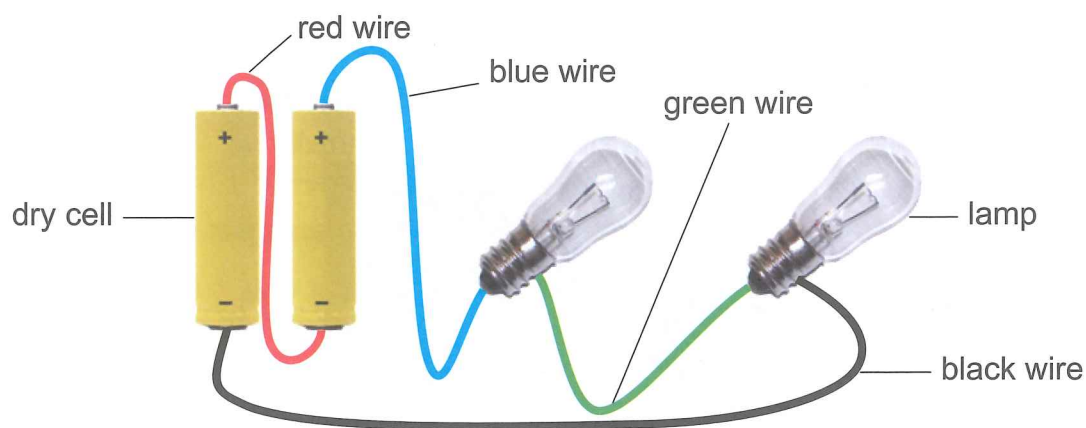


(D)



5. Mark made an electrical circuit using two dry cells and two lamps. He also used four wires, each of which had a different colour plastic coating.

The picture shows the circuit that he made.



What is the colour of the wire that connects the two lamps to each other?

(A) red

(B) blue

(C) black

(D) green

6. Stan collected some rocks on his last holiday.

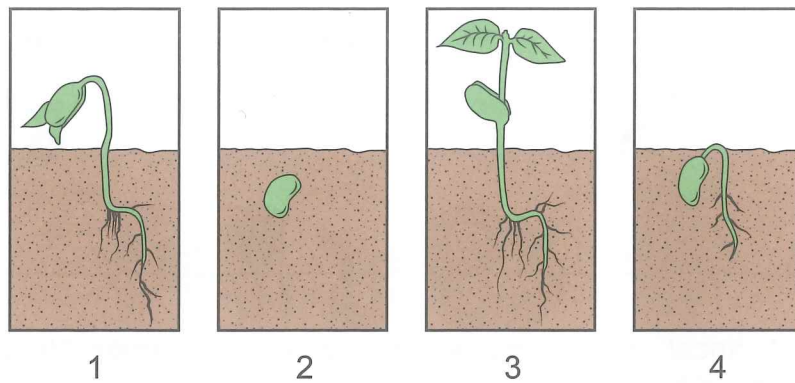
His favourite rock is the one that is smooth, rounded and has a light stripe.

Which of these rocks is Stan's favourite?



7. Irene has some pictures of a bean plant growing from a seed.

The pictures are not shown in the correct order of the seed growing.



What is the correct order?

- (A) 1, 4, 3, 2 (B) 2, 4, 1, 3 (C) 3, 1, 4, 2 (D) 4, 1, 3, 2

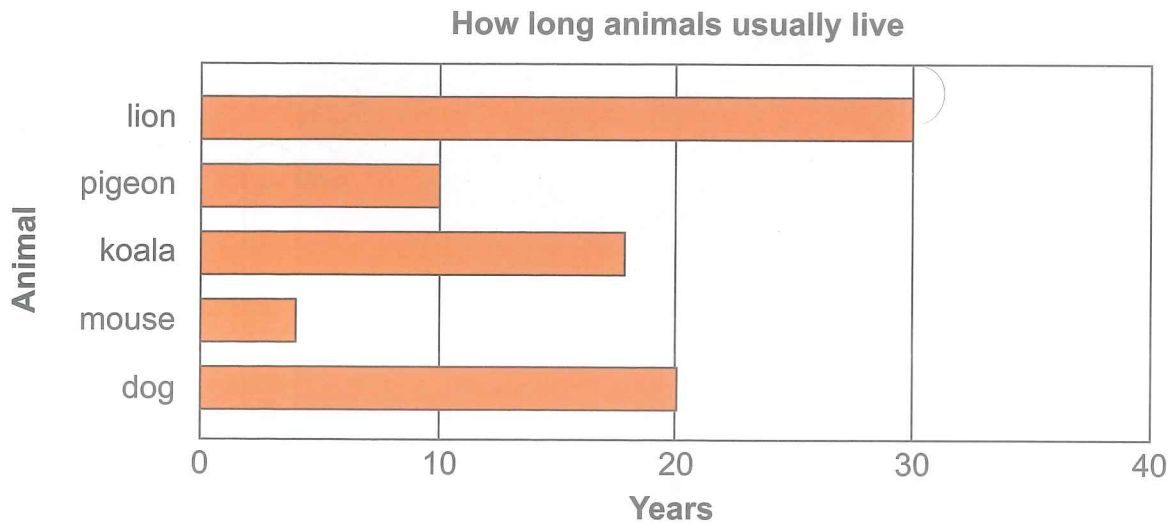
8. The diagram represents some craters on the surface of the Moon.



Which of these craters is the widest?

- (A) Q (B) R (C) S (D) T

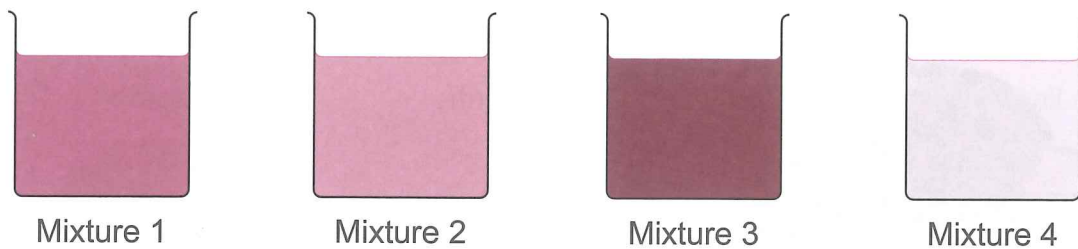
9. The bar graph shows how long five different animals usually live.



Using the information in the bar graph, which of these is true?

- (A) Dogs live longer than lions.
 - (B) Mice live longer than pigeons.
 - (C) Koalas live longer than pigeons.
 - (D) Dogs live longer than all the other animals.
10. Emma made four mixtures by adding different amounts of food colouring to water. She used the same type of food colouring in every mixture.

The diagrams show the four mixtures she made.

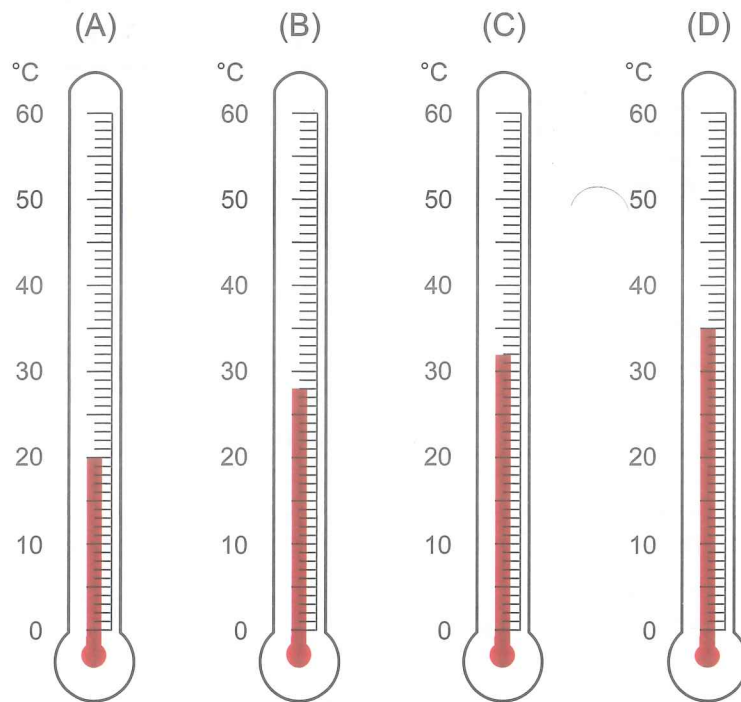


Which mixture has the greatest amount of food colouring in it?

- (A) Mixture 1 (B) Mixture 2 (C) Mixture 3 (D) Mixture 4

11. Adam has four thermometers.

Which thermometer shows a temperature of 32 °C?



12. The photograph shows a cicada next to a ruler.

Cicadas have two sets of wings. The longer wings are called forewings.



How long is the forewing of this cicada?

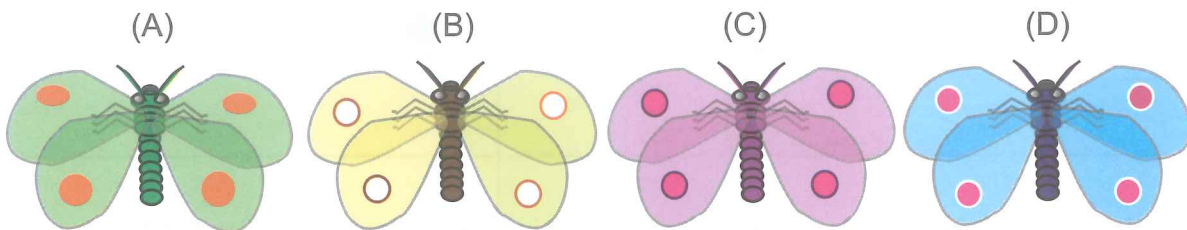
- (A) 3 cm
- (B) 6 cm
- (C) 7 cm
- (D) 9 cm

13. The photograph shows some flowers that Henry has growing in his garden.

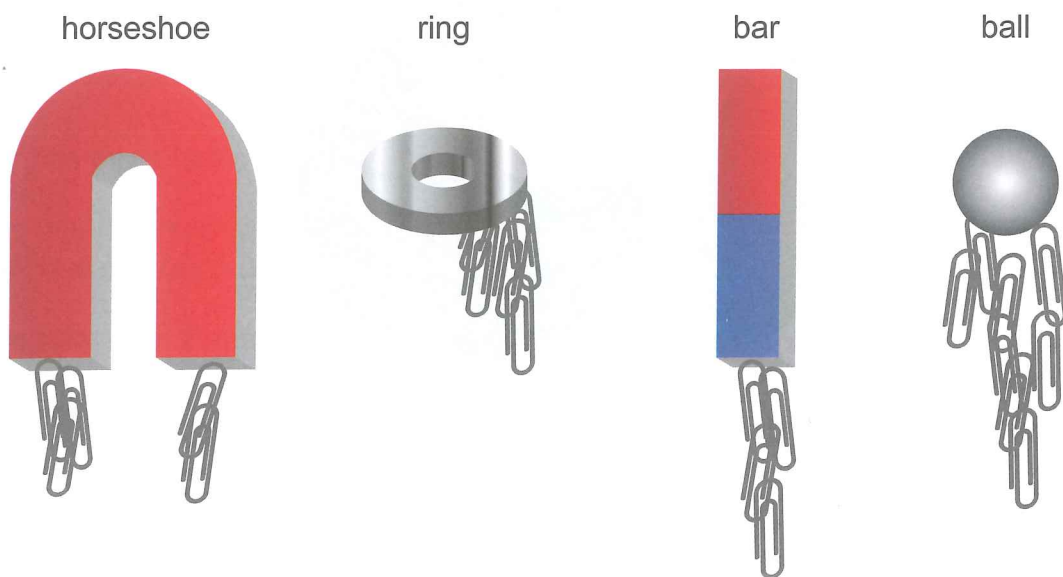


He found butterflies with different colours in his garden.

Which butterfly would be most difficult to see if it rests on these flowers?



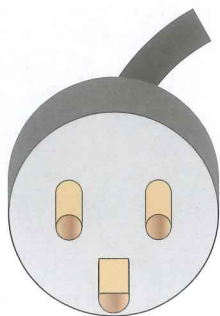
14. Ned has four different magnets. He knows that magnets can pick up small iron objects. He used each magnet to pick up as many iron paperclips as it could.



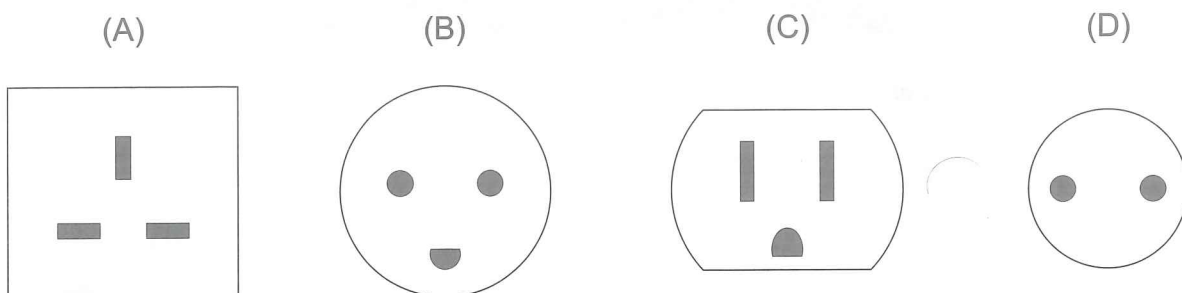
Which is the strongest magnet?

- (A) ball (B) bar (C) ring (D) horseshoe

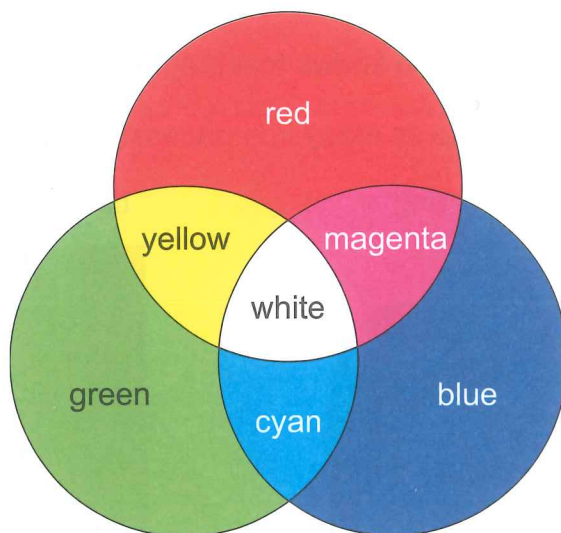
15. Many electrical appliances have to be plugged into a power outlet. This diagram shows the plug of an electrical appliance.



Into which power outlet will this plug fit?



16. The diagram shows what happens when you shine red, green and blue light onto white paper.



What colour light is produced where red light overlaps only with green light?

- (A) cyan (B) magenta (C) yellow (D) white

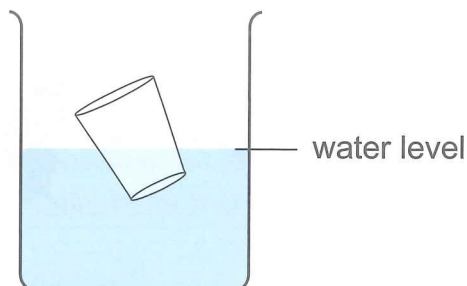
17. On Monday morning Joe washed his sweatshirt and hung it outside. It took 4 hours to dry. On Wednesday morning Joe washed the same sweatshirt again and hung it outside. It took 2 hours to dry.



Compared to the weather on Monday, the weather on Wednesday was most likely

- (A) more rainy.
- (B) drier and more windy.
- (C) colder and more cloudy.
- (D) more cloudy with no wind.

18. Lisa placed an empty plastic cup in a jar of water. The plastic cup floated in the water.



What would happen to the water level in the jar if Lisa placed a marble in the plastic cup?

- (A) The water level would be lower.
- (B) There would be more water in the jar.
- (C) The water level would move upwards.
- (D) The water level would not change.

19. Lily has three rocks – rock 1, rock 2 and rock 3.

She wants to see which rock is the hardest. She rubs rock 1 and rock 2 together. Rock 2 now has a scratch on it.

She rubs rock 2 and rock 3 together. Rock 3 now has a scratch on it.

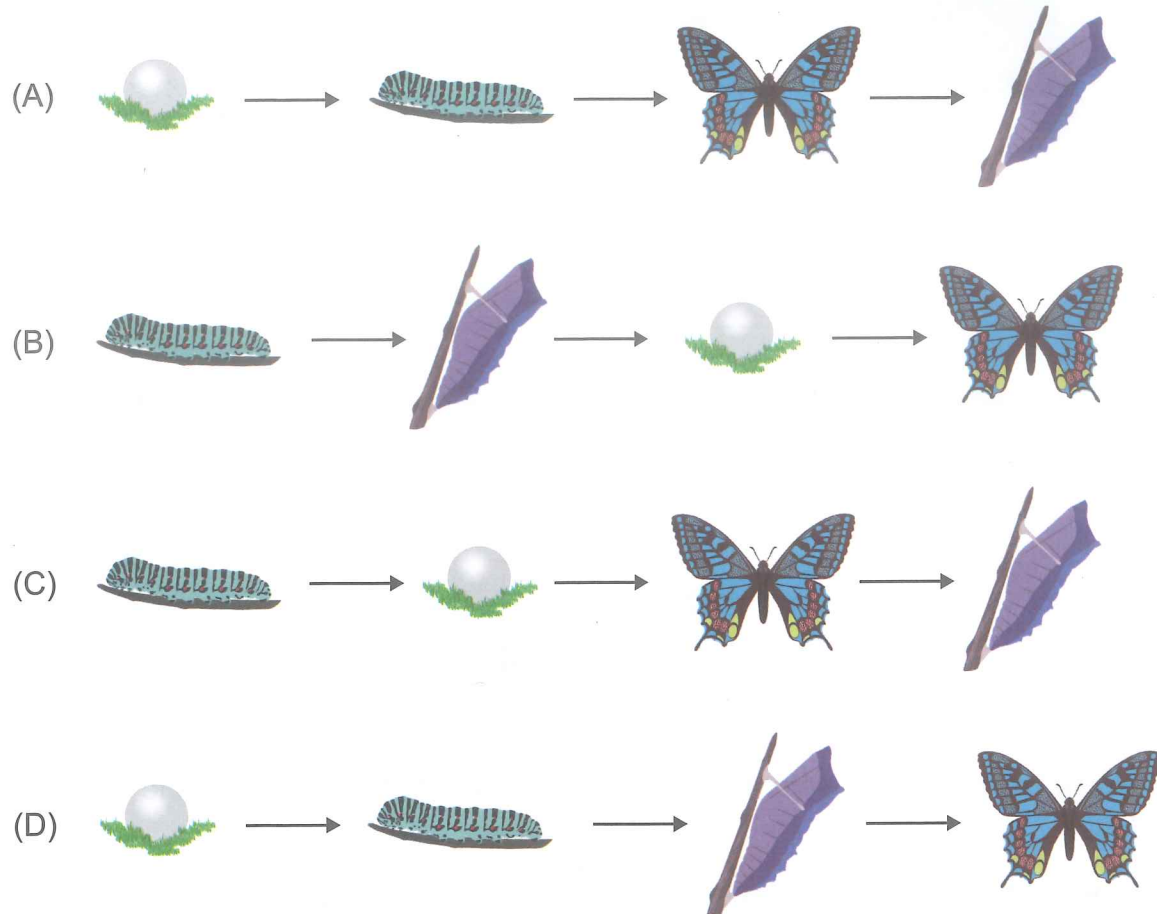
Which rock is the hardest?

- (A) rock 1
- (B) rock 2
- (C) rock 3
- (D) All rocks have the same hardness.

20. The life cycle of a butterfly has four stages. A female butterfly lays eggs on a leaf. The eggs hatch and caterpillars crawl out. The caterpillars eat leaves and grow larger.

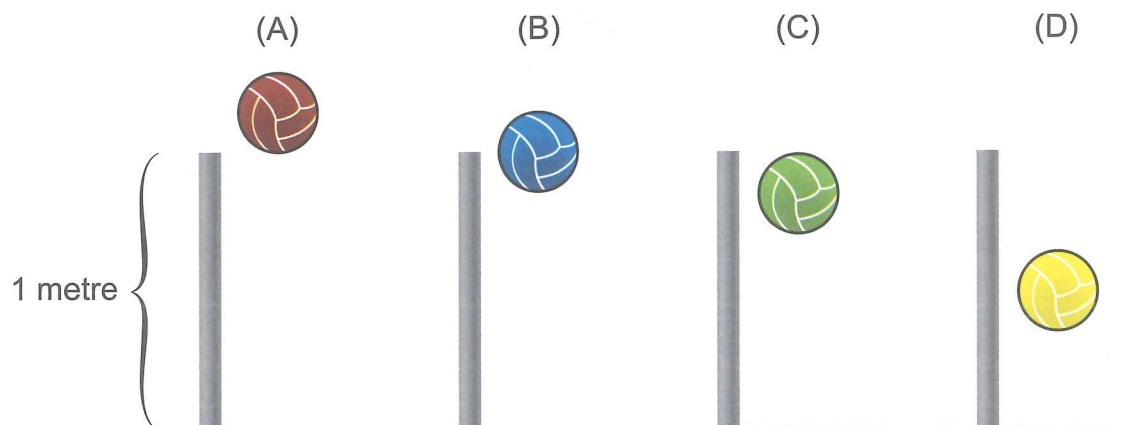
Each caterpillar spins a cocoon around itself. After a while, a butterfly comes out of each cocoon.

Which sequence shows the life cycle of a butterfly?

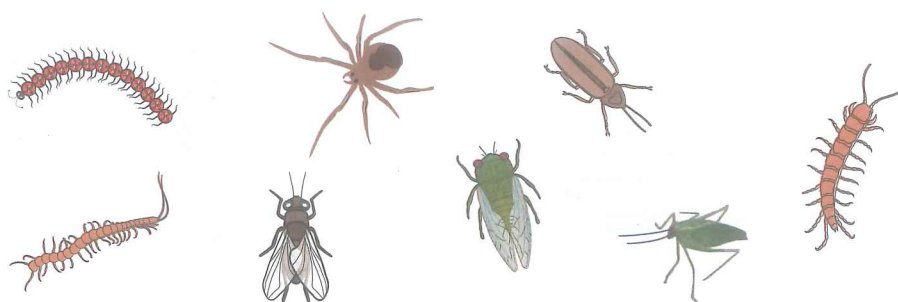


21. Maree wants to test how well different balls bounce by dropping them all from the same height. She will drop the balls so they all fall one metre.

Which diagram shows a ball at the correct height to begin her experiment?



22. The picture shows eight animals found in a garden.



The table lists some features of the animals in the picture.

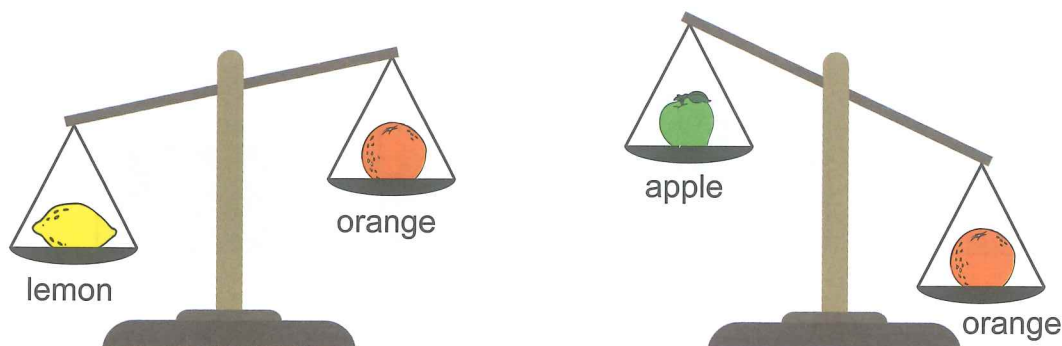
Animal	Feature
spider	• 8 legs • no antennae on head
insect	• 6 legs • 2 antennae on head
centipede	• more than 8 legs • 2 antennae on head • 1 pair of legs on each body part
millipede	• more than 8 legs • 2 antennae on head • 2 pairs of legs on each body part

How many insects are there in the picture?

- (A) 4 (B) 3 (C) 2 (D) 1

23. Sam had a lemon, an orange and an apple. He used a set of scales to work out which was the heaviest.

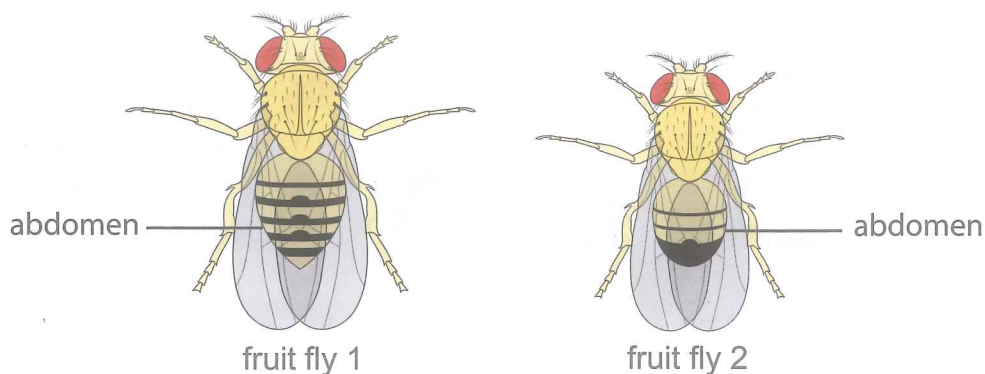
The diagram shows his results.



Which fruit was the heaviest?

- (A) lemon (B) orange
(C) apple (D) The orange and apple were the heaviest.

24. This picture shows two fruit flies.



There are differences between the body of a male fruit fly and that of a female fruit fly.

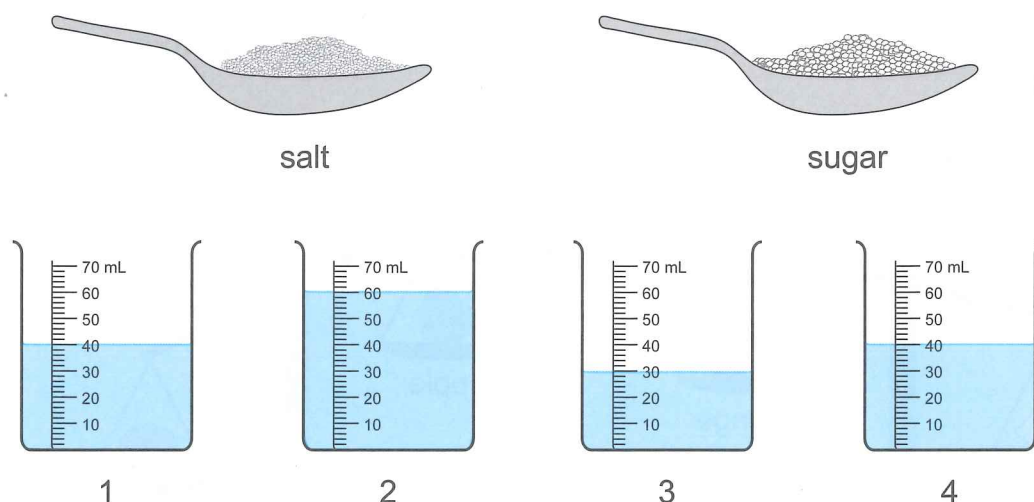
- Males have a round abdomen, females have a pointed abdomen.
- Females are usually larger than males.
- The last section of the abdomen in a male is darker than that of a female.

Which sentence about these two fruit flies is correct?

- (A) They are both male.
- (B) They are both female.
- (C) Fruit fly 1 is a male and fruit fly 2 is a female.
- (D) Fruit fly 1 is a female and fruit fly 2 is a male.

25. Eve wants to see which dissolves faster in water, salt or sugar.

She takes a teaspoon of salt and a teaspoon of sugar. Both teaspoons are the same size. Each amount of salt and sugar is the same.



She has four containers of water, but wants to use two containers.

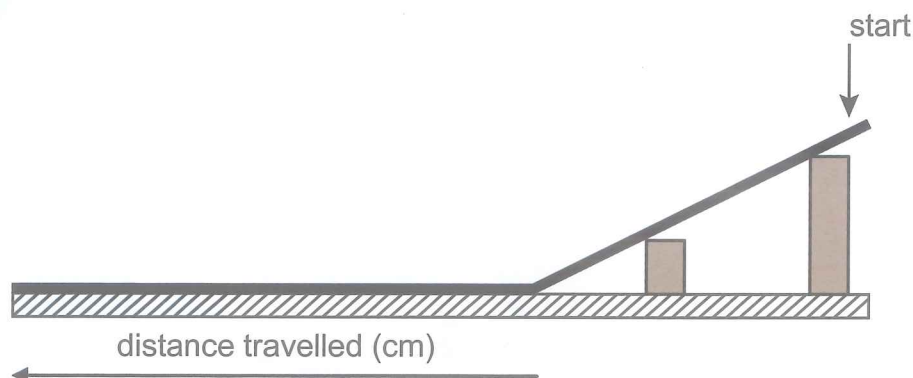
Which two containers of water must Eve use?

- (A) 1 and 2
- (B) 2 and 3
- (C) 3 and 4
- (D) 1 and 4

For questions 26 and 27 use the information below.

Lisa has four different-sized marbles. She wants to see if the size of the marble is the only thing that affects how far it travels when it rolls down a ramp.

The diagram shows how she set up her experiment.



The table gives the size and mass of her marbles.

				
	marble 1	marble 2	marble 3	marble 4
size of marble (cm)	$\frac{1}{2}$	1	$1\frac{1}{2}$	2
mass of marble (gram)	5	12	12	25

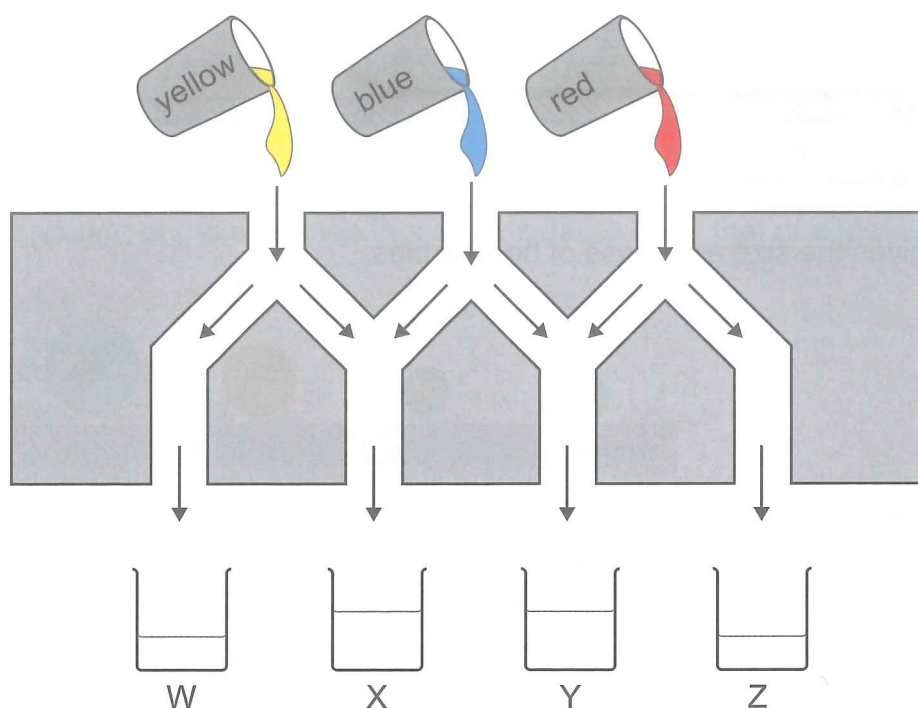
26. Which two marbles should Lisa use to check whether size affects how far it travels?
- (A) 1 and 2
(B) 2 and 3
(C) 3 and 4
(D) 4 and 1
27. What can Lisa do to make the marble roll faster?
- (A) decrease the height of the ramp
(B) increase the height of the ramp
(C) make the distance travelled longer
(D) make the distance travelled shorter

28. Eric and Rena have three tins of paint, one yellow, one blue and one red.

They know that:

- mixing yellow paint and blue paint makes green paint;
- mixing red paint and blue paint makes purple paint;
- mixing red paint and yellow paint makes orange paint.

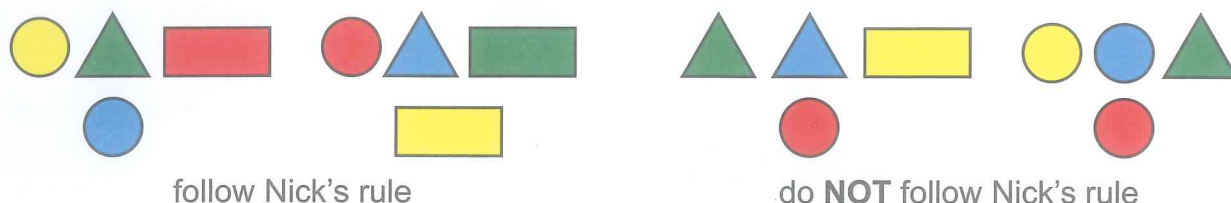
They pour the same amount of each paint into a special mixing machine and collect the paints in four containers: W, X, Y and Z.



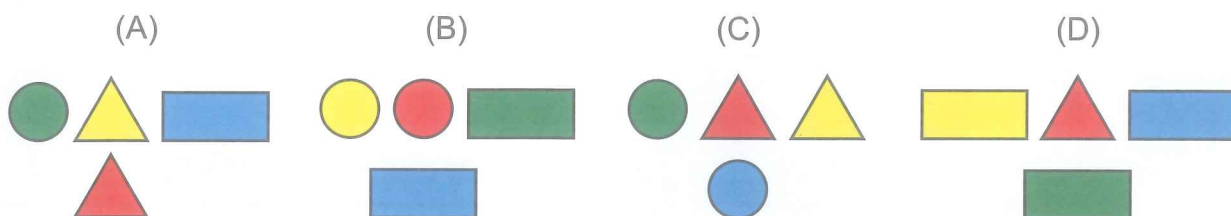
What is the colour of the paint in each container?

	W	X	Y	Z
(A)	green	purple	blue	red
(B)	green	blue	purple	red
(C)	yellow	blue	red	purple
(D)	yellow	green	purple	red

29. Nick was playing with coloured tiles. He put tiles in groups according to a rule. The picture shows two groups that follow Nick's rule and two groups that don't.

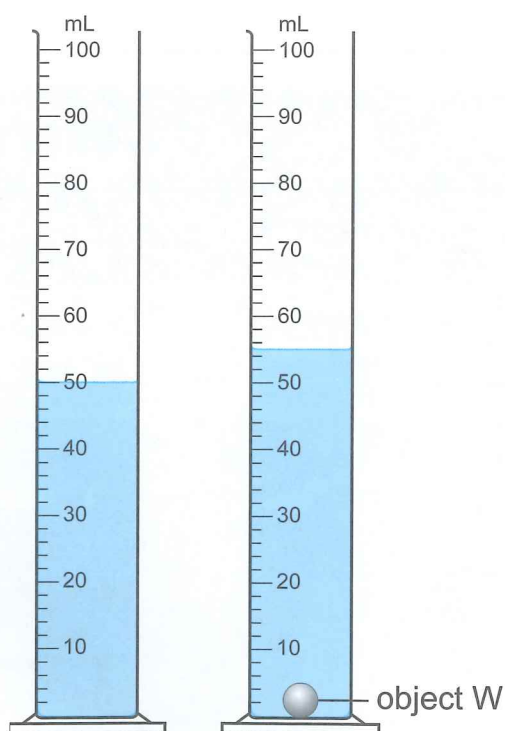


Which of these groups has been arranged according to Nick's rule?



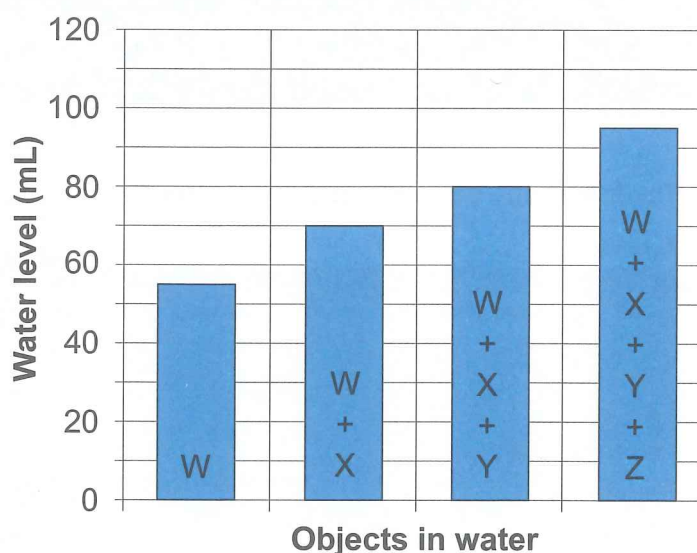
30. Ted has four objects: W, X, Y and Z.

Ted adds 50 millilitres (mL) of water to a measuring cylinder. He places object W into the water and records the new water level.



He then adds each object in turn and records the new water level.

The column graph shows his results.



Which two objects have the same volume?

- (A) W and X (B) X and Y (C) X and Z (D) Y and Z